

TrES-3b

R-2805

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140717 · created 2026-09-05T00:41:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456855.7539 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.055
Transit depth (Rp/Rs)^2	2.15 +/- 1.6 [%]
Orbital Inclination (inc)	82.68 +/- 2.58
Airmass coefficient 1 (a1)	1.002 +/- 0.019
Airmass coefficient 2 (a2)	0.0 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#3 - [570, 342]
Optimal Aperture	3.44
Optimal Annulus	5.73
Transit Duration (day)	0.053 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.055 vs 0.16309 (deviation 0.24σ)
Duration fit vs published	0.053 d vs 0.0567 d (deviation 0.12σ)
Mid-time O–C	-3.6 min
Depth SNR	2.2
Residual scatter	1.93 %

Light curve

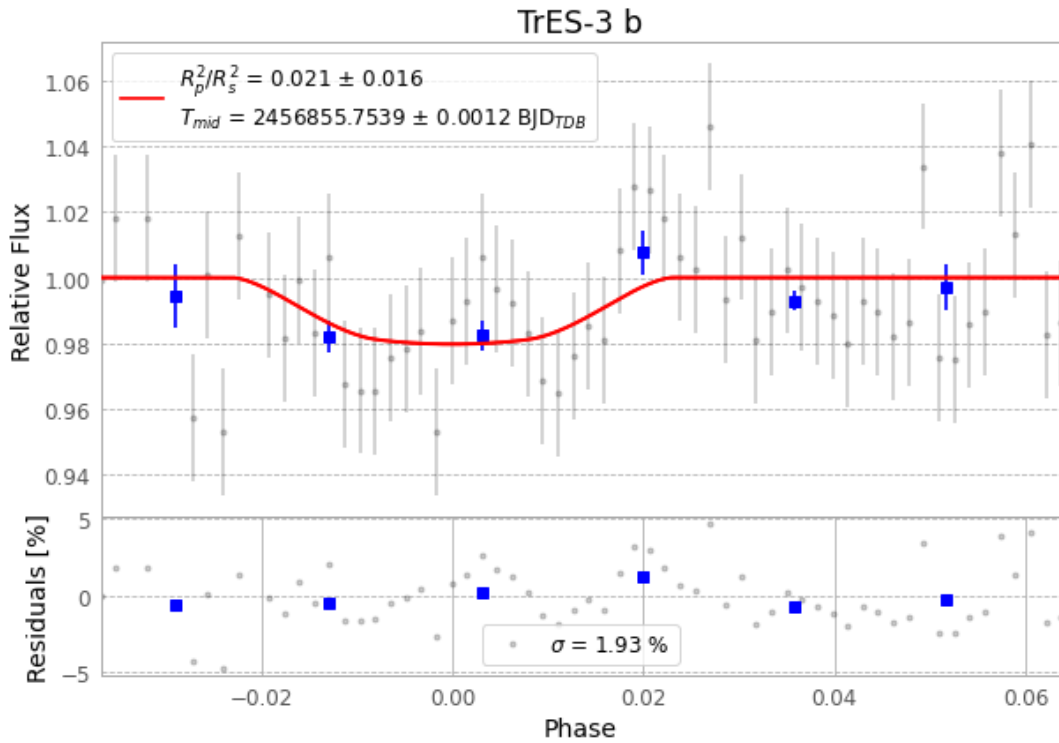


Figure 1 — FinalLightCurve_TrES-3 b_2014-07-16.png (SHA-256 190f1f014a094f3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [277, 327], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 63cf4447ecc752db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), TRES-3140717041812.FITS → TRES-3140717080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140717.log (ab67d069d999...), FinalLightCurve_TrES-3 b_2014-07-16.png (190f1f014a09...), FinalLightCurve_TrES-3 b_2014-07-16.csv (c8acaa9c1e60...)

Compute resources

Wall time 958.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:41Z.